

Diwaakar Jayaprakash

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SUMMARY

Mechanical Engineering undergraduate with hands-on experience in CAD-driven design, thermal analysis, and simulation. Proficient in SolidWorks, Fusion 360, ANSYS (thermal/structural), MATLAB, and Python. Experienced in additive manufacturing and rapid prototyping. Seeking internship roles in energy-efficient systems, automation, and data center infrastructure.

EDUCATION

IIITDM Kancheepuram — B.Tech in Mechanical Engineering

2023 – Present

Relevant Coursework: Heat Transfer, Manufacturing Processes, Kinematics & Design of Machines, Materials Science

EXPERIENCE

Engineering Intern — Tech Mahindra (Client: ASML)

May 2025 – Jul 2025

Semiconductor Manufacturing Engineering

Bangalore, India

- Operated high-throughput semiconductor equipment (1,000+ outputs/min); developed a MATLAB/Python pipeline for CPD analysis with real-time dashboards for process monitoring and defect detection.
- Performed volumetric constraint analysis in Siemens NX; delivered a technical presentation on semiconductor manufacturing processes.

Research Assistant — EV Safety & Battery Systems

Mar 2025 – Apr 2025

Research Internship under Dr. Raja, HOD of Mechanical, IIITDM Kancheepuram

- Conducted thermal analysis of Li-ion battery circuits using ANSYS Thermal and infrared thermal imaging under varied load conditions for a research publication; assisted in prototype manufacturing and testing.

PROJECTS

Rover Development — Team Shunya — Mars Research Station

Apr 2024 – Present Multi-year

Mechanical Design Engineer

Competition Project

- International Rover Challenge (IRC) 2025 — Goa, India — Ranked 16th / 226 teams globally**
 - ScienceCacheSystem:** Designed a compact autonomous sample-collection assembly integrating a 6-beaker carousel (100 ml each), helical-screw drilling actuator, and chemical-mixing module within a single actuator-minimized structure. Produced full SolidWorks CAD models with BOMs; fabricated via FDM 3D printing (PLA); resolved critical dust-sealing challenges at mechanism interfaces.
 - Modular Chassis:** Engineered a lightweight chassis using PVC pipes and 3-way joints, optimized for rapid assembly, center-of-mass management, and integration with suspension, differential, and battery systems. Validated suspension kinematics and obstacle clearance via SolidWorks Motion Analysis.
 - Led on-site assembly, integration testing, and maintenance of all rover subsystems, including a 6-DoF manipulator arm and electromechanical interfaces.
- European Rover Challenge (ERC) Remote 2025 — Poland — Ranked 4th / 50+ teams globally**
 - Authored competition-grade technical documentation and design reports covering design intent, testing protocols, and subsystem specifications. Managed outreach, sponsorship acquisition, branding, and social media campaigns.
- International Rover Challenge (IRC) 2026 — Udipi, India — Ranked 10th / 200+ teams globally**
 - Contributed to technical documentation, design validation reports, and competition deliverables; continued team management responsibilities.

Key Skills: Autodesk Fusion 360 (Design, assembly), SolidWorks (Motion Analysis), ANSYS Structural FEA, FDM Manufacturing, System Integration, Technical Documentation

Smart Transparent Window System — Energy & Automation-Oriented Prototype

Jan 2025 – Mar 2025

Independent Research Project

IIITDM Kancheepuram

- Developed a 4-layer multifunctional smart window integrating a fiberglass substrate, electrochromic film, transparent perovskite solar panel, and transparent LED display within a 3D-printed housing. Performed power generation calculations and feasibility analysis for energy-efficient building applications.

Key Skills: Multifunctional Material Integration, Energy Systems Analysis, Rapid Prototyping, Design Under Constraints

TECHNICAL SKILLS

- Design & CAD:** SolidWorks (Design & Motion Analysis), Autodesk Fusion 360, Siemens NX, AutoCAD, Rhinoceros 3D, OnShape
- Simulation & Analysis:** ANSYS (Thermal), Basic FEA, CFD Fundamentals
- Programming & Data:** MATLAB, Python, MySQL, C++
- Manufacturing & Prototyping:** 3D Printing (FDM, Resin), CNC Machining, Laser Cutting
- Engineering Domains:** Mechanical System Integration, Thermal Systems, Energy-Efficient Design, Automation-Oriented Design

AWARDS & ACHIEVEMENTS

- 10th Globally,** International Rover Challenge (IRC) 2026
- 4th Globally,** European Rover Challenge (ERC) Remote 2025
- 16th Globally,** International Rover Challenge (IRC) 2025 (out of 226 teams)